

Experiment No. 1

Title: Determination of Metacentric height

Objective: To determine the metacentric height of a ship model

Apparatus& Accessories Used:

- 1) Ship model Apparatus
- 2) Weights

Introduction:

Metacentre is defined the point about which the submerged body oscillates when it is subjected to small angular tilt. It can also be defined as the point at which the line of action of the force of buoyancy will meet the normal axis of the body when the body is given a small angular displacement.

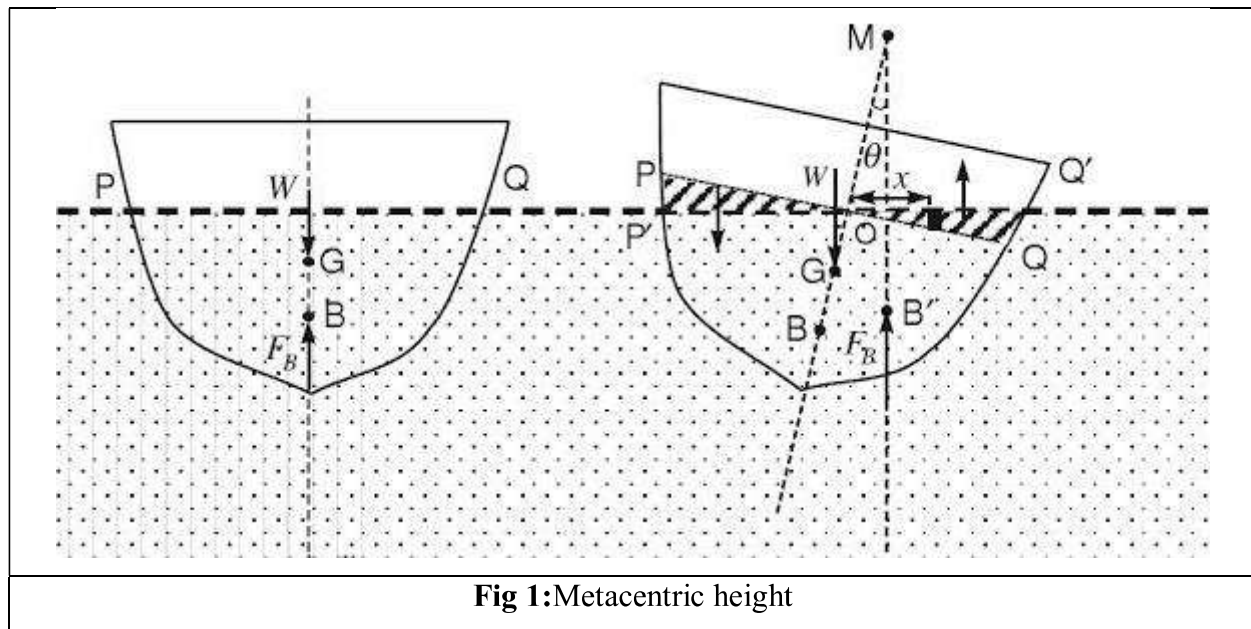


Fig 1: Metacentric height

Stability of Floating Bodies

A body is in a stable equilibrium if it returns to its original position after being slightly displaced. Neutral position if the object remains in the new position after being slightly displaced. A body is in an unstable equilibrium if it continues to move in the direction of the displacement

If the Centre of Buoyancy (B) is defined as the centre of gravity of the displaced fluid then the stability of a floating object will depend on whether a righting or overturning moment is developed when the centre of gravity (G) and the centre of buoyancy move out of vertical alignment due to the shifting of the position of the later. The centre of buoyancy moves because if a floating body tips, the shape of the displaced liquid changes.

Let **W** be the weight of the vessel including W_1 (applied load),

G be the Centre of gravity of the vessel,

B be the Centre of Buoyancy of the vessel and

M be the Metacentre of the vessel.

The distance between **G** and **M**, i.e., **GM** is known as the *Metacentric Height*.

The weight W_1 is moved across the vessel towards right through a distance 'x'. The vessel will be tilted. The angle of heel θ is measured by means of a plumb line and a protractor attached on the vessel. The new centre of gravity of the vessel will shift to G_1 as the weight has been moved towards right. Also the centre of buoyancy will change to B_1 as the vessel has tilted. Under equilibrium the moment caused by the movement of the load W_1 through a distance 'x' must be equal to the moment caused by the shift of the centre of gravity from G to G_1 .

Thus, the moment due to the change of G

$$\begin{aligned} &= W \times GG_1 \\ &= W \times GM \tan\theta \end{aligned}$$

The moment due to moment of W_1 ,

$$\begin{aligned} &= W_1 \times X \\ \Rightarrow W_1 \times X &= W \times GM \tan\theta \end{aligned}$$

Therefore,

$$\Rightarrow GM = \frac{W_1 \times X}{W \times \tan\theta}$$

Description of the Apparatus:

In this apparatus a tank of size 65cm x 55cm x 42cm is used where a ship model comprises of a Perspex sheet scale, a protector, a pointer etc. are applied to determine the metacentric height.

Experimental Procedure:

1. Fill the tank about 2/3 full of water and note down the initial level of water with the scale.
2. Put the ship model in the tank and note down the level of water again.

3. Knowing the cross-sectional area of tank and increase (rise) in the level of water, the weight of the water displaced by the vessel can be obtained, which the same as the weight of ship model is W.
4. Adjust the pointer to zero reading for the initial condition when no tilting movement is applied to the vessel.
5. Apply a tilting movement to the vessel with the help of moveable weight.
6. Corresponding to applying tilting movement note the reading of pointer indicating the angle of heel θ .
7. Take the readings of θ corresponding to different distances of the moveable weight.
8. Calculate the metacentric height.

Observation Tables:

Sl. No.	Initial level of water, $H_1(\text{m})$	Final level of water, $H_2(\text{m})$	Rise in level of water, $H=H_2-H_1(\text{m})$	Weight placed on ship, $W_1(\text{N})$	Distance at which weight is placed, $X(\text{m})$	Angle of heel, θ (degree)

Calculations:

Area of tank (to be calculated), $A = \underline{\hspace{2cm}} \text{ m}^2$

Density of water, $\rho = 1000 \text{ kg/m}^3$

Weight of ship model, $W = \rho \times g \times A \times H \text{ (N)}$

Metacentric height, $GM = \frac{W_1 \times X}{W \times \tan \theta} \text{ (m)}$

Conclusion:

The mean metacentric height is =

Experiment No. 2

Title: Verification of Bernoulli's Theorem

Objective: To determine the summation of total head i.e. pressure head, velocity head (or kinetic head) and the potential head (or datum head) are constant for an incompressible, irrotational, in-viscid and laminar flow

Apparatus Used: Bernoulli's Apparatus

Introduction:

Considering frictionless flow along a variable area duct, the law of conservation of energy states that for an in-viscid, incompressible, irrotational and steady flow along a stream line the total energy (or head) remains the same. This is called Bernoulli's equation. The total head of flowing fluid consists of pressure head, velocity head and elevation head.

Hence,

$$\frac{P_1}{w_1} + \frac{V_1^2}{2g} + z_1 = \frac{P_1}{w_2} + \frac{V_2^2}{2g} + z_2 = \text{constant}$$

here, P, V and Z refer to the pressure, velocity and position of the liquid relative to some datum at any section. Real liquids are viscous and flows are always accompanied by though in some cases they may be neglected.

Description of the Apparatus:

The apparatus consists of a hydraulic bench where supply water tank, discharge measuring tank, variable area duct with minimum area at the middle and connections to piezometer tubes at different positions of the duct, stop watch, meter rod etc. are attached. The cross-sectional area of the duct at the piezometer location, area of the volumetric tank etc. is to be measured.



Fig 1: Bernoulli's Apparatus

Experimental Procedure:

1. Measure the distance of each piezometer tube from the inlet section and area of cross-section.
2. Measure the area of discharge measuring tank.
3. Adjust the flow when the head in tube becomes constant. Note the height in all tubes i.e. pressure head $\frac{P}{w}$
4. Note the initial reading of water measuring tank and collect water for certain time 't' in tank and note final reading of water. Calculate the discharge. By dividing the discharge with area of cross-section under each tube calculate the velocity head $\frac{V^2}{2g}$. As the duct is horizontal so the height of each tube from ground is constant. Head 'z' is zero.
5. Calculate the total head $\frac{P}{w} + \frac{V^2}{2g} + z = \text{constant}$
6. Repeat the experiment for different flows.

Observation Table 1:

Run no.	Initial level, $h_1(\text{m})$	Final level, $h_2(\text{m})$	Time, t (s)	Discharge $Q = \frac{(h_2 - h_1)A}{t} (\text{m}^3/\text{s})$

Observation Table 2:

Tube No.	1	2	3	4	5	6	7	8	9
Distance from inlet section(m)									
Area of cross section of duct, a x b (m ²)									
Velocity $V = \frac{Q}{a \times b}$ (m/s)									
Velocity Head = $\frac{V^2}{2g}$ (m)									
Datum Head Z=0									
Pressure Head $= \frac{P}{w} + Z(m)$									
Total head (m)									

Calculations:

Width of duct (b) = _____ m

Area of collecting tank A = _____ m²

Total Head, $\frac{P}{w} + \frac{V^2}{2g} + z = \text{constant (m)}$

where, w = ρg

$$\frac{P_1}{w_1} + \frac{V_1^2}{2g} + z_1 = \frac{P_1}{w_2} + \frac{V_2^2}{2g} + z_2$$

(i.e. sum of total heads at sec.1= sum of total heads at sec.2)

Conclusion:

The total head for each tube and the significance of the total head with the corresponding tubes (i.e. 1&9, 2&8 etc.) are as follows:

Experiment No. 3

Title: Determination of C_v , C_c and C_d for an orifice

Objective: To determine the hydraulic co-efficient (C_v , C_c and C_d) for an orifice

ApparatusUsed: Hydraulic coefficients of orifice Apparatus

Introduction:

Orifices are devices used for discharging fluids in to the atmosphere. It is the opening in the wall of a tank or in a plate which may be fitted in a pipe such that the plate is normal to the axis of the pipe. The discharging fluid from the tank/conduit through the orifice comes out in the form of a free jet. In the process, the total energy of the fluid in the tank is converted to kinetic energy as the jet issues out in to the atmosphere. The jet cross-section initially contracts to a minima and then expands partly due to the viscous resistance offered by the surrounding atmosphere and partly due to inertia of the fluid particles. The section which has the minimum area of cross-section is known as vena-contracta. The contraction and expansion of the jet results in loss of energy.

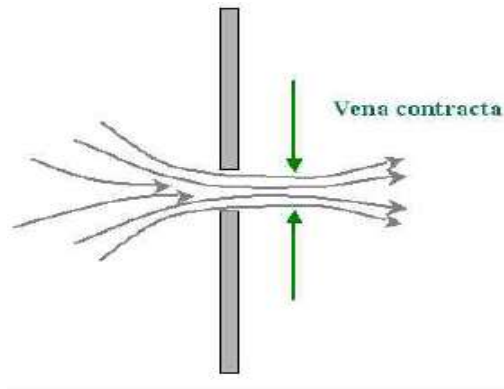


Fig 1: Schematic diagram of vena contracta

The actual velocity at vena-contracta is smaller than the theoretical velocity due to frictional resistance at the orifice edges. The ratio between the actual velocity and the theoretical velocity of the jet is known as co-efficient of velocity, C_v . Its value depends upon the size and shape of the orifice and head causing flow. C_v for a vertical orifice is determined experimentally by measuring the horizontal and vertical co-ordinates of the issuing jet. The water flows through an orifice under a constant head H . Theoretical velocity = $\sqrt{2gH}$.

Let V be the actual velocity (which is horizontal) of the jet. While covering horizontal distance, the jet is acted upon by gravity with a downward acceleration 'g'. Consider a small particle of water at vena-contracta and suppose it falls through a vertical distance 'y' in a horizontal distance 'x' in time 't' sec.

Then,

$$\begin{aligned}x &= Vt \\y &= \frac{1}{2}gt^2 \\ \Rightarrow y &= \frac{1}{2}g \frac{x^2}{V^2} \\ \Rightarrow V &= \sqrt{\frac{gx^2}{2y}} \\ \therefore C_v &= \frac{\text{Actual velocity}}{\text{Theoretical velocity}} \\ &= \frac{\sqrt{\frac{gx^2}{2y}}}{\sqrt{2gH}} \\ &= \frac{x}{\sqrt{4yH}}\end{aligned}$$

The ratio between the actual discharge and the theoretical discharge is called the Coefficient of Discharge, C_d .

$$\begin{aligned}\therefore C_d &= \frac{\text{Actual discharge}}{\text{Theoretical discharge}} \\ &= \frac{Q_{\text{act}}}{Q_{\text{th}}} \\ &= \frac{\frac{(h_2 - h_1)A}{t}}{a\sqrt{2gH}}\end{aligned}$$

Now, we know that,

$$\begin{aligned}C_d &= C_c \times C_v \\ \Rightarrow C_c &= \frac{C_d}{C_v}\end{aligned}$$

Description of the apparatus:

The apparatus consists of a hydraulic bench where a reservoir, inlet tank, collecting tank, volumetric tank and a motor of 0.5HP are mounted. In the inlet tank outside wall an orifice of dia. 1 cm is mounted. Finally the discharge through orifice is to be measured.



Fig 2:Hydraulic coefficients of orifice Apparatus

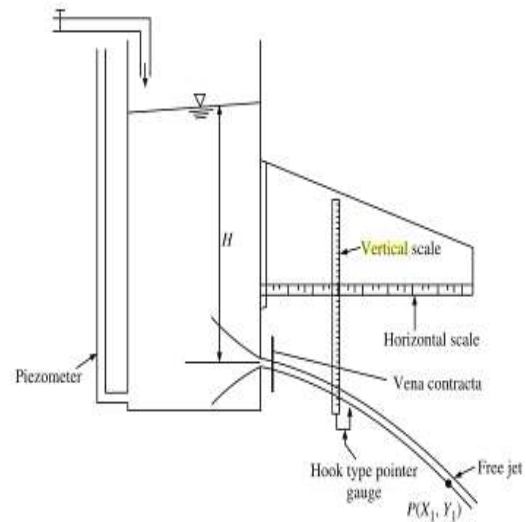


Fig 3:Schematic diagram of the apparatus

Experimental Procedure:

- i. The diameter of the orifice, dimension of measuring tank and diameter of the pipeline were noted.
- ii. The X and Y movement of the pointer was checked to be jerk free and smooth.
- iii. The inlet controlled valve was opened. The inlet tank was allowed to fill the overflow started. The inlet valve was adjusted till the water level in the tank becomes constant as checked by piezometer reading.
- iv. After 5 to 10 mins., when steady of flow acquired it and valve of X and Y were measured open pointer was kept at center of jet.
- v. The discharge was measured and head 'H' was calculated again. The procedure was repeated for a set of readings.

Observation Table:

Sl. No.	Measuring tank reading			Head, H (m)	x = x ₂ -x ₁ (m)	y = y ₂ -y ₁ (m)
	Initial level, h ₁ (m)	Final level, h ₂ (m)	Time, t (s)			

Calculations:

Diameter of circular orifice = 0.01 m

Area of circular orifice, $a = \frac{\pi d^2}{4} = 0.00785 \text{ m}^2$

Area of collecting tank, $A = \text{_____} \text{ m}^2$

Actual Discharge, $Q_{\text{act}} = \frac{(h_2 - h_1)A}{t} \text{ (m}^3/\text{s)}$

Theoretical Discharge, $Q_{\text{th}} = a \sqrt{2gH} \text{ (m}^3/\text{s)}$

Coefficient of Discharge, $C_d = \frac{Q_{\text{act}}}{Q_{\text{th}}}$

Co-efficient of velocity, $C_v = \frac{x}{\sqrt{4yH}}$

Co-efficient of Contraction, $C_c = \frac{C_d}{C_v}$

Conclusion:

The standard values for different hydraulic co-efficient are

Note: The values for different hydraulic co-efficient should be

$$C_d = 0.60 \text{ to } 0.65$$

$$C_d = 0.61 \text{ to } 0.69$$

$$C_d = 0.95 \text{ to } 0.99$$

Experiment No. 4

Title: Study of the flow over notch (Triangular and Rectangular notch) and to find out the coefficient of discharge

Objective: To study the flow over a notch and to find the co-efficient of discharge for it along with to calibrate it for discharge measurement in a free surface flow

Apparatus Used: Hydraulic bench of flow over notches (Triangular or Rectangular) with a fitted water tunnel

Introduction:

A weir is an obstruction placed across a free-surface flow such that the flow takes place over it. Notches are openings cut in metallic plates and installed in flumes or small channels. Installation of a notch is exclusively for the purpose of measuring the discharge in the system. A sharp crested weir or notch for the measurement of discharge generally have a regular geometrical shape like triangular. The free surface flow taking place over it acquires steady state conditions such that the discharge is uniquely related to the head 'H' over the crest of the notch measured at a distance about 3 to 4 times from the crest towards upstream. The value of co-efficient of discharge 'C_d' is given by the ratio $\frac{Q_{ac}}{Q_{th}}$.

Description of the apparatus:

A tank fitted with a notch perforated plates, A discharge measuring tank with a piezometer tube and a graduated scale, Triangular notch of varying angles, A stop watch, A hook gauge with a vernier scale etc. are fitted to the hydraulic bench.



Fig 1: Apparatus for discharge measurement using notches

Experimental Procedure:

1. Fix the plate having Rectangular notch or V- notch in the water passage of Hydraulic bench.
2. Turn the hydraulic bench on; water will accumulate in the channel.
3. When the water level reaches the Crest or sill of notch stop the inflow and note the reading, and design it as H_1 .
4. Restart the bench and note the volume and time of water that accumulates in the volumetric tank of bench, from this find the discharge, and also note the height of water at this point H_2 .
5. Find $H = H_2 - H_1$ this will give you the head over the notch.
6. Find the width of the notch.
7. Take different readings by changing the discharge head over the notch, using the above procedure.
8. Plot a graph between H and Q and find K from graph equation.

Observation Table:

Sl. No.	Initial water level above notch, $H_1(m)$	Final water level above notch, $H_2(m)$	Measuring of tank reading		
			Initial level of tank, $h_1(m)$	Final level of tank, $h_2(m)$	Time, t (s)

Calculations:

Area of measuring tank,

$$A = \text{_____} \text{ m}^2$$

Angle of V notch,

$$\theta = 60^\circ$$

Width of rectangular notch,

$$B = 0.125 \times 0.15 = 0.01875 \text{ m}^2$$

Head over notch,

$$H = H_2 - H_1 \text{ (m)}$$

Actual Discharge,

$$Q_{\text{act}} = \frac{(h_2 - h_1)A}{t} \text{ (m}^3/\text{s)}$$

Theoretical Discharge for V notch,

$$Q_{\text{th}} = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{\frac{5}{2}} \text{ (m}^3/\text{s)}$$

Theoretical Discharge for rectangular Notch,

$$Q_{\text{th}} = \frac{2}{3} C_d 2g B H^{\frac{3}{2}} \text{ (m}^3/\text{s)}$$

Coefficient of Discharge,

$$C_d = \frac{Q_{\text{ac}}}{Q_{\text{th}}}$$

In actual practice, the discharge over a notch is considerably less than indicated by the above formula without considering ' C_d ' i.e. the formula is derived on the basis of frictionless one dimensional flow. The discrepancy arises due to real flow effects like viscosity, end contractions, nappe suppression, ventilation of weirs etc. so the actual discharge is obtained by multiplying the theoretical discharge by ' C_d ' as given in the above formula.

Conclusion:

Experiment No. 5

Title: Free vortex flow

Objective Of the Experiment: To determine the velocity of rotating fluid mass

Apparatus Used: Free vortex flow apparatus

Introduction:

A free vortex flow exists when no external torque to rotate the fluid mass. The fluid rotates by virtue of some rotation previously imparted to it. The streamline are then concentric circles and there is no variation of total energy with radius. i.e. $V \times r = \text{const.}$

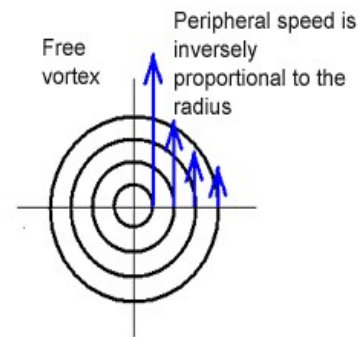


Fig 1: Schematic Diagram

Description of the apparatus:

In this apparatus Perspex pipe of diameter 300mm is used. Four circumferential jets have been place along the circumference of cylinder. A hole near its bottom centre which helps in formation of the free vortex. A pointer gauge is used to measure the co-ordinate of vortex at various points. The supply of water is received to hydraulic bench.



Fig 2: Free Vortex Apparatus

Experimental Procedure:

1. Locate the centre point of the cylindrical vessel with the help of scale attached to the apparatus.
2. Switch on the pump/ motor
3. Allow the water to pass through the nozzles placed at the circumference of the cylinder.
4. Control the flow of water with the help of valve control key.
5. Wait for sometime till the vortex is formed.
6. Consider a point on the free surface near the starting of the vortex and note its vertical co-ordinate as y_0 .
7. The co-ordinates of a point (r,y) on the free surface of the vortex are noted. (Where r is the horizontal co-ordinate and y is the corresponding vertical co-ordinate)
8. Repeat the process for more points on the free surface.

Observation Table:

Sl.No.	r(m)	y(m)	$C = \sqrt{(y_0 - y)2gr^2}(\text{m}^2/\text{sec})$	$V = \frac{C}{r}(\text{m/sec})$

Calculations:

$$y = y_0 - \frac{C^2}{2gr^2}$$

$$\Rightarrow \frac{C^2}{2gr^2} = (y_0 - y)$$

$$\Rightarrow C^2 = (y_0 - y)2gr^2$$

$$\Rightarrow C^2 = \sqrt{(y_0 - y)2gr^2}$$

The velocities at the different points are calculated using the relation,

$$V = \frac{C}{r}$$

Conclusion:

Experiment No. 6

Title: Forced vortex flow

Objective: To determine the velocities at different points on a forced vortex for different angular velocities

Apparatus Used: Forced Vortex flow apparatus

Introduction:

A forced vortex manifests when the fluid elements are made to rotate in concentric circles with constant angular velocity ω . The velocity distribution is prescribed by the relation $V = r\omega$ i.e., the velocity grows with an increase in distance from the vortex centre. The static pressure difference between any two points, say, 1 and 2, in a forced vortex flow is given as:

On the free surface of the vortex, $P_1 = P_2 = \text{Atmospheric pressure}$

$$P_1 - P_2 = \frac{\rho\omega^2}{2}(r_2^2 - r_1^2) + \rho g(y_1 - y_2)$$

$$\because P_1 = P_2$$

$$y_2 - y_1 = \frac{\omega^2}{2g}(r_2^2 - r_1^2)$$

$$\Rightarrow \omega^2 = \frac{2g(y_2 - y_1)}{r_2^2 - r_1^2}$$

$$\Rightarrow \omega = \sqrt{\frac{2g(y_2 - y_1)}{r_2^2 - r_1^2}}$$

where, (r_1, y_1) and (r_2, y_2) are the co-ordinates of 1 and 2 considering the centre of the base of the cylinder as the origin.

Description of the apparatus:

This apparatus is used to visualize the forced vortex. In this apparatus a cylindrical Perspex pipe of diameter 300 mm is used. A DC motor of 0.5 H.P. with thyristor control is used to

rotate the cylinder. A point gauge (depth gauge) is used to measure the co-ordinates of vortex at various points.



Fig 1: Forced Vortex Apparatus

Experimental Procedure:

1. Locate the centre point of the cylindrical vessel with the help of a pointer/ depth gauge scale attached to the apparatus.
 2. Bring down the pointer gauge up to the bottom centre of the jar and note down the reading on the vertical gauge, termed it as Y_0
 3. The cylinder of the Perspex jar is initially filled with water up to its half mark.
 4. Switch on the variable speed motor and control the speed with the regulator attached to it.
 5. Allow the water to rotate through the cylinder.
 6. Allow the water to rotate for some time and have a look on the surface profile.
(as in fig.8.2).
 7. Wait till the vortex is formed and stabilized.
 8. Consider any two points on the free surface one at near the centre and the other towards the periphery of the rotating fluid mass.
 9. Note these two points as r_1 , r_2 and the corresponding y_1 , y_2 .
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10. The actual value of these co-ordinates should be calculated with reference to the centre point and the datum line of the apparatus.
11. Repeat the process for other co-ordinate points with different speed of the motor.

Observation Table:

Sl. No.	y_1 (m)	y_2 (m)	r_1 (m)	r_2 (m)	$\omega = \sqrt{\frac{2g(y_2 - y_1)}{r_2^2 - r_1^2}}$ (rad/s)	$V_1 = r_1\omega$ (m/s)	$V_2 = r_2\omega$ (m/s)

Conclusion:

The flow velocities at different points on the free surface of the vortex are thus evaluated by considering different speeds of rotation.

Experiment No

Title: Study of hydraulic force.

Objective: To Study the effect of force on following types of vanes: Hemispherical Vane
& Flat Plate Vanes.

Apparatus Used: Impact of Jet apparatus.

Introduction:

When a plate is placed in the path of a jet, the jet exerts a force on the plate. This force can be calculated from the impulse momentum equation. Momentum equation is based on Newton's Second law of motion, which states that "The algebraic sum of external forces applied to control volume of fluid in any direction is equal to the rate of change of momentum in that direction." The external forces include the components of the weight of the fluid and of the forces exerted externally upon the boundary surface of the control volume.

THEORY:

If a vertical water jet moving with velocity 'V' is made to strike a target, which is free to move in the vertical direction, then a force will be exerted on the target by the impact of jet. According to momentum equation, this force (which is also equal to the force required to bring back the target in its original position) must be equal to the rate of change of momentum of the jet flow in the same direction.

Due to impact of the jet on the flat stationary plate, the entire velocity of the jet is destroyed and due to the rate of change of momentum, force acts on the plate. The jet after striking will move along the plate. But the plate is at right angles to the jet. Hence the components of the velocity of the jet in the direction of the jet after striking it will be zero.

F , = Rate of change of momentum in the direction of force

$$\begin{aligned} F &= \frac{\text{Initial momentum} - \text{Final momentum}}{\text{time}} \\ &= \frac{\text{Mass} \times \text{initial velocity} - \text{Mass} \times \text{Final velocity}}{\text{time}} \\ &= \frac{\text{Mass}}{\text{time}} \times (\text{initial velocity} - \text{final velocity}) \\ &= \rho A V (V - 0) \\ &= \rho A V^2 \text{ (for Flat Vane)} \end{aligned}$$

$$F_X = 2 \rho A V^2 \text{ (for Hemispherical Vane)}$$

DESCRIPTION:

The setup consists of a sump tank with centrifugal pump to circulate water. A chamber with side glass is provided for visualization of impact of jet on vanes. Water from sump tank flows through a nozzle and strikes vertically to vane positioned above the nozzle. Two types of vanes (Hemispherical /flat) are provided that can be fixed one at a time. Arrangement is made for the movement of the plate of the vane under the action of the jet and also because of the weight placed on the loading pan. Measuring tank and stop watch is provided for flow measurement.

UTILITIES REQUIRED:

- Electricity Supply: Single Phase, 220 V AC, 50 Hz, 5-15 Amp. Combined socket with earth connection.
- Water Supply (Initial Fill).
- Floor Drain Required.
- Floor Area Required: 1.5 m x 0.7 M.

EXPERIMENTAL PROCEDURE:

- STARTING PROCEDURE:
 1. Close all the valves provided V_1 to V_4
 2. Fill sump tank $\frac{3}{4}$ with water and ensure that no foreign particles are there.
 3. Fix desired vane (Hemispherical /Flat).
 4. Ensure that all On/Off switches given on the panel are at OFF position.
 5. Open by pass valve V_2 .
 6. Switch on the pump.
 7. Put weight on the pan.
 8. Operate by pass valve V_2 and flow control valve V_1 to regulate the flow of water through channel.
 9. Now control the flow of water so that the applied weight on the top is counter balanced by the impact of jet (Pan leaves its rest position on the spring).
 10. Measure flow rate using measuring tank and stop watch.
 11. Repeat the experiment for different weights.
 12. Repeat the experiment for other vane.
- CLOSING PROCEDURE:
 1. When experiment is over, switch off pump.
 2. Switch off power supply to panel.
 3. Drain water from all tanks with the help of drain valves V_4 & V_3 .

OBSERVATION:

DATA:	
Acceleration due to gravity $g = 9.81 \text{ m/sec}^2$	Weight of aluminum disc with rod $W_{D+R} = 0.213 \text{ kg}$
Diameter of nozzle $d = 0.01 \text{ m}$	Weight of Flat plate vane $W_F = 0.025 \text{ kg}$
Area of measuring tank $A = 0.077 \text{ m}^2$	Weight of Hemispherical vane $W_H = 0.044 \text{ kg}$
Density of water $\rho = 1000 \text{ kg/m}^3$	

OBSERVATION TABLE

For Flat Plate					For Hemispherical Plate				
Sl No	W _A (kg)	R ₁ (cm)	R ₂ (cm)	t (sec)	Sl No	W _A (kg)	R ₁ (cm)	R ₂ (cm)	t (sec)
1					1				
2					2				
3					3				
4					4				
5					5				

CALCULATION:

$$a = \frac{\pi}{4} d^2 (m^2)$$

$$R = \frac{R_1 - R_2}{100} (m)$$

$$Q = \frac{A \times R}{t} (m^3/sec)$$

$$V = \frac{Q}{A} (m/sec)$$

$$F_x = \rho A V^2 (N) \text{ (for flat vane)}$$

$$F_x = 2 \rho A V^2 (N) \text{ (for hemispherical vane)}$$

$$W = W_{D+R} + W_F + W_A (kg) \text{ for flat plate}$$

$$W = W_{D+R} + W_H + W_A (kg) \text{ for hemispherical plate}$$

$$F_{TH} = W \times g (N)$$

$$\text{Error} = \frac{F_x - F_{TH}}{F_x} \times 100\%$$

CONCLUSION: